

MCS-EXP-08(code)

1) Lead_Bode plot

```
clc;
clear;
close all;
%% Plant
G = tf(1,[1 4 3]);
%% Required additional phase
phi_required = 35; % degrees
%% Convert to radians
phi = deg2rad(phi_required);
%% Calculate alpha
alpha = (1-sin(phi))/(1+sin(phi));
fprintf('alpha = %.5f\n',alpha);
%% Choose desired crossover frequency
wc = 2; % rad/s
%% Calculate T
T = 1/(wc*sqrt(alpha));
fprintf('T = %.5f sec\n',T);
%% Lead compensator
Clead0 = tf([T 1],[alpha*T 1]);
%% Calculate gain so that |CG| = 1 at wc
mag = abs(squeeze(freqresp(Clead0*G,wc)));
K = 1/mag;
fprintf('Lead gain K = %.5f\n',K);
%% Final compensator
Clead = K*Clead0;
%% Compensated open loop
Llead = Clead*G;
%% Bode plot
figure;
margin(Llead);
grid on;
title('Bode Plot - Lead Compensated System');
%% Closed-loop system
Tlead = feedback(Llead,1);
%% Step response
figure;
step(Tlead);
grid on;
title('Lead Compensated Closed-Loop Response');
```

2) Lead_Root locus

```
clc;
clear;
close all;
%% Plant
G = tf(1,[1 4 3]);
%% Root Locus
figure;
rlocus(G);
grid on;
title('Root Locus of the System');
%% Choose desired gain
K = 10;
%% Open-loop system with gain
L = K*G;
%% Closed-loop system
T = feedback(L,1);
%% Display closed-loop poles
poles = pole(T);
fprintf('Chosen gain K = %.2f\n',K);
fprintf('Closed-loop poles:\n');
disp(poles);
%% Step response
figure;
step(T);
grid on;
title('Closed-Loop Step Response');
```

